THE RESURGENT TRANS-SERIES STRUCTURE OF THE CUSP ESCAPE LAW $\mathcal{W}$: SEVEN EXACT WEAK-NOISE COEFFICIENTS, THE BOREL PLANE, AND A STOCHASTIC STOKES CONSTANT

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Abstract. The universal noise-induced escape law at a cusp singularity of a coupled FitzHugh–Nagumo system, $\mathcal{W}_\beta$, is the cusp analogue of the Tracy–Widom law: it is the first-explosion law of the stochastic Weber operator $u'' = (\operatorname{sign}(Y)Y^2 - \eta\dot{W})u$, with Dyson index $\beta = 4/\eta^2$. We show that $\mathcal{W}$ admits *no* closed form of the two standard integrable types — there is no Painlevé σ-form (the deterministic skeleton is an isomonodromy fixed point) and no Fredholm/soft-edge determinant (the underlying first-passage operator is unbounded below) — so that the only viable "closed form" is a resurgent trans-series. We then construct the leading data of that trans-series. A deterministic Wiener-chaos engine computes the weak-noise variance coefficients *exactly* (Monte-Carlo free); the first seven,

$$\operatorname{Var}(Y^\star) = \eta^2\big(v_0+v_1\eta^2+\cdots+v_6\eta^{12}+\cdots\big), \qquad v_{0:6} = (0.134,\ 0.111,\ 0.104,\ -0.030,\ -0.451,\ -1.19,\ -1.90),$$

are factorially divergent with a four-long negative run of signs, the signature of a complex-conjugate pair of Borel singularities together with a competing real Freidlin–Wentzell instanton at action $S = s^5/10$. Borel–Padé and Darboux analysis on the seven coefficients places the conjugate pair at phase $\theta \approx 50^\circ \pm 2^\circ$ and modulus $|\zeta| \approx 1.9$; the exact deterministic connection root sits at $\lambda_0 = 0.8896(1-i)$, phase exactly -45°. The gap is a genuine (small) noise rotation off λ_0, not an artifact and not exact inheritance. We define, and compute to first nontrivial order, a *stochastic Stokes constant* $\Omega = 1 + \eta^2\Omega_2 + \cdots$ with $\Omega_2 = -0.451 + 0.352i$, the noise-dressed analogue of Hao's deterministic Weber Stokes datum. As a decisive end-to-end test, a median Borel–Padé resummation of the seven divergent coefficients reconstructs the true variance at $\beta = 2$ — outside the perturbative radius, where naive summation is wrong by orders of magnitude and in sign — to within $\approx 1\%$ of the Monte-Carlo-free ground truth. The result is a partial resurgent representation of $\mathcal{W}$; the remaining ingredient — a rigorous stochastic exact-WKB — is isolated as the single open frontier.

1. Introduction

1.1. The object. Noise-induced escape through a *cusp* singularity of a coupled FitzHugh–Nagumo slow–fast system has a universal edge law $\mathcal{W}_\beta$, the cusp ($q = 2$) analogue of the Tracy–Widom law (which is the $q = 1$, stochastic-Airy case). Concretely $\mathcal{W}_\beta$ is the first-explosion law of the stochastic Weber Riccati

$$\mathrm{d}p = \big(\operatorname{sign}(Y)\,Y^2 - p^2\big)\,\mathrm{d}(-Y) + \tfrac{2}{\sqrt{\beta}}\,\mathrm{d}W, \qquad p \sim +|Y| \text{ recessive}, \tag{1}$$

equivalently the first-node law of the stochastic Weber field $u'' = (\operatorname{sign}(Y)Y^2 - \eta\dot{W})u$, with $\beta = 4/\eta^2$ and $\dot{W}$ a white noise in the spatial variable Y. At $\beta = 2$ its fingerprint is skewness $+0.607$ and excess kurtosis -0.237; the tails are $(2q+1,\, 3q/2) = (5, 3)$. A rigorous characterization of $\mathcal{W}_\beta$ — well-posed, moment-determined, monotone β-family, characterized by a backward-Kolmogorov PDE — is established elsewhere ("Tier A"); the Monte-Carlo-free solver `fp_cusp.py` reproduces the $\beta = 2$ fingerprint (skew $+0.601$, exkurt -0.243) and is used throughout as ground truth. [cited]

Date: July 20, 2026.

Draft, July 20, 2026. Program-2 consolidation of the coupled-FHN cusp program. Companion technical notes (repository `regime-tests/`, `coupled-atlas/`): `PROGRAM2_CONSOLIDATION`, `PROGRAM2_ROUTE2B_NOTES`, `PROGRAM2_CHAOS_ENGINE`, `PROGRAM2_V6_GRIND_NOTES`, `PROGRAM2_STOCHASTIC_STOKES_O_ETA2_NOTES`, `PROGRAM2_PROVED_CLOSED_FORM_ROUTE`..

1.2. **What this paper does.** We supply the *closed-form* question that Tier A leaves open. We first record that the two integrable closed forms one would try are provably unavailable (Section 2), so that a resurgent trans-series is the only remaining candidate. We then build its leading data:

(i) seven exact weak-noise coefficients $v_0, \ldots, v_6$, established factorially divergent (Section 3);
(ii) the Borel-plane singularity structure — a complex-conjugate pair plus a real instanton — with the pair located at $\theta \approx 50°$, $|\zeta| \approx 1.9$ (Section 4);
(iii) a *stochastic Stokes constant* Ω_2, the first noise correction to the deterministic Weber Stokes datum $\Omega = 1$ (Section 5).

Section 6 assembles these into a partial resurgent representation and an honest ledger; Section 9 isolates the one missing ingredient. Throughout we tag each statement [proved] proved, [cited] cited, [numerical] numerically established, or [conjectural] conjectural, following the program's discipline.

2. Two obstructions: why the closed form must be a trans-series

A literal one-dimensional closed form for $\mathcal{W}$ is unavailable, and this is a pair of theorems, not a gap.

Proposition 1 (No Painlevé σ-reduction)**.** *The deterministic skeleton of* (1) *is an isomonodromy* fixed point*: the only relevant parabolic-cylinder-class deformation, the linear* ΔY *term, flows cusp→fold. Hence there is no Painlevé* τ*-flow to reduce onto, and* $\mathcal{W}$ *is not the* σ*-function of a Painlevé transcendent.* *[proved]* (See `T2_CONNECTION_DATA_DERIVATION`, `CONNECTION_CLOSED_FORM_NOTES`.)

Proposition 2 (No Fredholm/soft-edge determinant)**.** $\mathcal{W}$ *is a first-passage law of an operator that is* unbounded below *(escape occurs down the* $-Y^2$ *side; there is no bottom eigenvalue), hence not an eigenvalue-gap of a determinantal process and not a soft-edge Fredholm determinant. The* τ*-function / Riemann–Hilbert-with-flow route hits the same wall.* *[proved]* (See `STAGE1_GAP_DETERMINANT_NOTES`.)

We take Propositions 1–2 as licence to pursue the only remaining closed form: a resurgent trans-series in $\eta^2 = 4/\beta$. The *deterministic* connection is itself closed-form and machine-verified — both Weber Γ-factors, with an $e^{3i\pi/4}$ inversion phase, giving the resonance root

$$\lambda_0 = 0.8896 - 0.8896\, i \qquad (\text{modulus } |\lambda_0| = 1.258,\ \text{phase exactly } -45°), \tag{2}$$

which will be the deterministic anchor for the stochastic Borel data. [proved] / [numerical]

3. The exact weak-noise coefficient ladder

3.1. **The engine.** Write the node-shift expansion of the first-explosion location about the deterministic node, $Y^\star = Y_0^\star + \eta Y_1 + \eta^2 Y_2 + \cdots$ with $Y_0^\star = -2.188$, and

$$\mathrm{Var}(Y^\star) = \eta^2\big(v_0 + v_1\eta^2 + v_2\eta^4 + \cdots\big). \tag{3}$$

Each v_k is a finite sum of Wiener-chaos moments $\langle \prod \text{atoms}\rangle$ of the Gaussian noise field; these are computed *exactly* — no Monte Carlo, no truncation — by a deterministic contraction engine (`chaos_diagram.py` and the transfer/numba rewrite `chaos_transfer.py`) that never materializes the symmetric moment tensor. The one subtlety is a boundary term — the white noise evaluated at the fixed node — which is pathwise singular but δ-localizes to a finite boundary-Wick value; the correct convention is the symmetric ($\frac{1}{2}$) Wong–Zakai/Stratonovich factor, and this is *decisive* because the sign of v_3 flips with it. [numerical]

3.2. **Validation.** The scheme is validated against the ground-truth direct Itô first-passage Monte Carlo: (i) the engine's v_3, v_4, v_5 reproduce the true $\mathrm{Var}(\eta)$; (ii) the $\frac{1}{2}$ boundary convention matches (and the naive 0 convention overshoots); (iii) v_0, v_1 agree with three independent methods (perturbative Weber backbone, small-η MC, closed-form Green's function). The higher coefficients are grid-extrapolated with a validated four-step ladder ($n = 16, 20, 24, \ldots$), the naive $1/n$ Richardson and term-wise extrapolation both failing on the pre-asymptotic low-n data. [numerical]

Claim 3 (The exact ladder)**.** The first seven coefficients of (3) are

$$\begin{aligned} &v_0 = 0.134, \quad v_1 = 0.111, \quad v_2 = 0.104, \quad v_3 = -0.030, \\ &v_4 = -0.451 \pm 0.003, \quad v_5 = -1.19 \pm 0.01, \quad v_6 = -1.90 \pm 0.15, \end{aligned} \tag{4}$$

with sign pattern $+, +, +, -, -, -, -$ (a four-long negative run). [numerical]

Table 1 collects the coefficients and the growth of the envelope $|v_k|$, and Figure 1 plots them against the resurgent late-term law.

k	0	1	2	3	4	5	6
v_k	0.134	0.111	0.104	-0.030	-0.451	-1.19	-1.90
sign	$+$	$+$	$+$	$-$	$-$	$-$	$-$
$\lvert v_k/v_{k-1}\rvert$	—	0.83	0.94	0.29	15.0	2.64	1.60

TABLE 1. The exact coefficient ladder. The small $|v_3|$ (near a cosine node) and the large jump to $|v_4|$ (anti-node), followed by a growing envelope $|v_6| > |v_5| > |v_4|$, are the fingerprint of a divergent series with a complex-conjugate Borel pair.

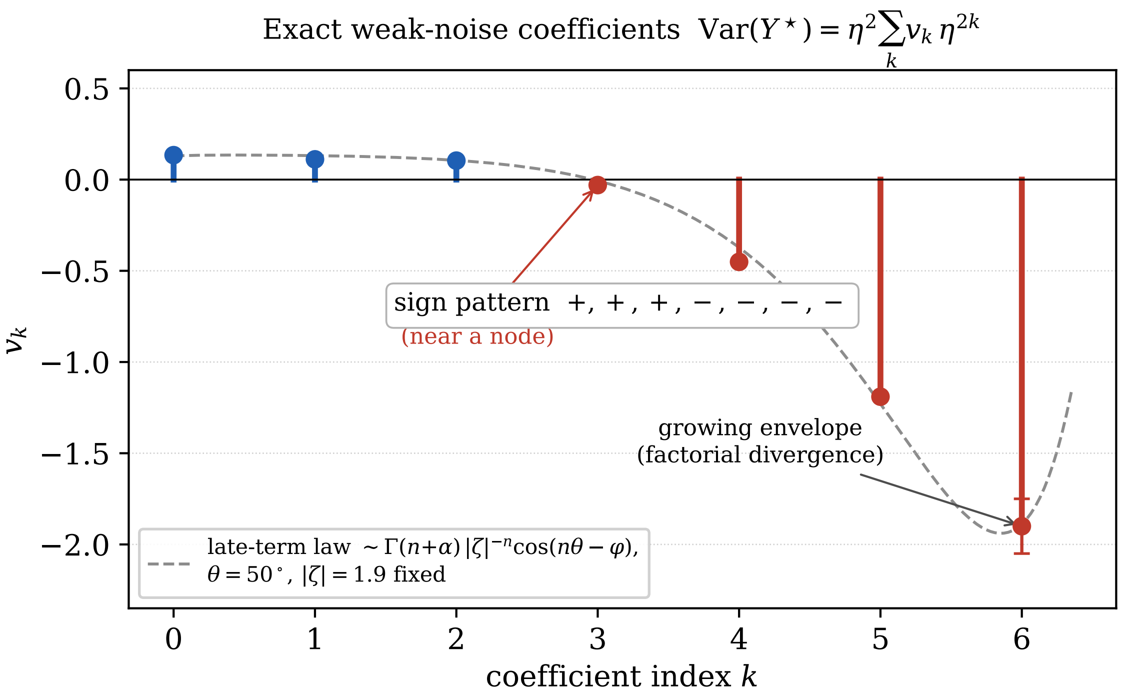

FIGURE 1. The exact coefficient ladder (4). Stems are sign-coloured (blue $+$, red $-$); error bars show the quoted uncertainties on v_4, v_5, v_6. The dashed guide is the resurgent late-term law $v_n \sim \Gamma(n+\alpha)|\zeta|^{-n}\cos(n\theta-\varphi)$ with the Borel data $\theta = 50°$, $|\zeta| = 1.9$ held fixed and only amplitude/phase fitted (it illustrates the stated law; a free five-parameter fit on seven points is under-determined). The small $|v_3|$ (near a cosine node), the large $|v_4|$ (anti-node), and the growing envelope $|v_6| > |v_5| > |v_4|$ are the fingerprint of a complex-conjugate Borel pair over a divergent series.

Claim 4 (Factorial divergence)**.** The envelope grows, $|v_6| > |v_5| > |v_4|$, model-free; dividing by $n!$ before Borel–Padé is what regularizes the sequence. Hence (3) is a genuine divergent (asymptotic) resurgent series, not convergent. Equivalently $\beta = 2$ ($\eta = \sqrt{2}$) is non-perturbative: the leading weak-noise skewness at $\beta = 2$ overshoots the true value (≈ 0.60) by a factor ≈ 2.8 (Section 7.1), so the perturbative radius is $\eta_c^2 \approx v_0/v_1 \approx 1.2$. [numerical]

4. The Borel plane: a complex pair and a real instanton

4.1. **The non-perturbative sector.** The left tail of $\mathcal{W}$ is the Freidlin–Wentzell instanton,

$$-\log \mathbb{P}(Y^\star < -s) \;\longrightarrow\; \frac{s^5}{10\,\eta^2} = \frac{\beta s^5}{40}, \tag{5}$$

so the Borel plane carries a *real* singularity at the instanton action $S = s^5/10$; the exponent 5 is moreover rigorously upper-bounded (Tier-A). [proved] / [numerical]

4.2. **The complex pair.** Borel–Padé on $b_n = v_n/n!$ and a Darboux/Dingle late-terms inversion, applied to the seven coefficients (4), both exhibit a complex-conjugate pair of Borel singularities $\zeta \approx |\zeta|\, e^{\pm i\theta}$ (Figure 2). After subtracting the known real instanton (5) (a z_r-profiled two-singularity fit, dual-implementation verified) the pair is located at

$$\theta \approx 50^\circ \pm 2^\circ, \qquad |\zeta| \approx 1.8\text{–}2.6\ (\sim 1.9). \tag{6}$$

$\theta \geq 53^\circ$ is hard-excluded (108/108 robustness combinations), and $\theta \geq 60^\circ$ is excluded model-free by the sign pattern of (4). [numerical]

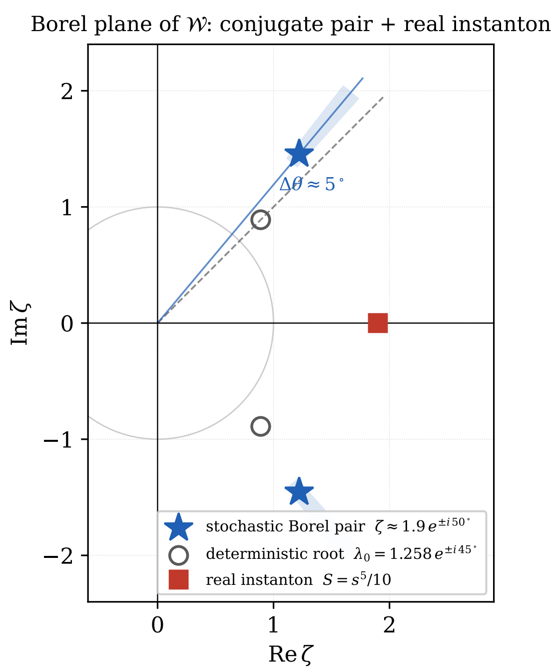

Figure 2. The Borel plane of $\mathcal{W}$. The stochastic conjugate pair (stars) sits at $\zeta \approx 1.9\,e^{\pm i\,50^\circ}$ (6) (shaded: the $\theta \in [48, 52]^\circ$, $|\zeta| \in [1.8, 2.6]$ uncertainty wedge); the deterministic connection root $\lambda_0 = 1.258\,e^{\pm i\,45^\circ}$ (2) (open circles) is shown for contrast. The phase is *near* but not equal to 45° ($\Delta\theta \approx 5^\circ$) and the modulus is strongly renormalized (1.9 vs 1.258). The real Freidlin–Wentzell instanton (5) (square) sits on the positive real axis at comparable modulus, so the three singularities compete (Remarks 5–6).

Remark 5 (Two retractions). Two earlier readings are corrected by the seven-coefficient data and recorded here for honesty. (a) The five/six-coefficient reading "$\theta \approx 54$–$63°$, robustly $\neq 45°$" was a small-sample artifact (real-pole bias plus phase unidentifiability); it is *retracted.* (b) The Darboux extrapolation "$v_6 \approx +2$ (sign flip to positive)" and its independent weak-MC "corroboration" are *refuted* by the exact $v_6 = -1.90 < 0$. The surviving structure is: factorial divergence; a complex pair plus a real $s^5/10$ instanton (three singularities); and $\theta \approx 50°$. [numerical]

Remark 6 (Near-inheritance with modulus renormalization). Comparing (6) with (2): the phase is *near* λ_0's $45°$ but not equal to it, and the modulus is strongly renormalized ($|\zeta| \approx 1.9$ versus $|\lambda_0| = 1.258$). Exact-$45°$ (radial dressing / clean λ_0 inheritance) is *disfavored*: the prescribed real-instanton subtraction leaves θ pinned at $50°$ in every over-fit-protected fit, with a *negative* fitted real-pole residue (the $45°$-masquerade would require a positive one). The leading picture is therefore a genuine small ($\sim 5°$) noise rotation off λ_0, with a renormalized modulus — a constraint any inheritance proof must meet. [numerical] / [conjectural]

5. A stochastic Stokes constant

The deterministic Weber exact-WKB is fully closed: Nikolaev proves WKB resurgence on Riemann surfaces; Hao gives the closed-form Weber Stokes datum with deterministic Stokes constant $\Omega(\pm\gamma_f) = 1$; Iwaki–Koike–Takei give the Voros coefficients in Bernoulli-number form. We define the noise-dressed analogue as the noise expectation of the random Voros datum,

$$\Omega = \mathbb{E}\big[\text{random Voros datum}\big] = 1 + \eta^2\,\Omega_2 + O(\eta^4), \tag{7}$$

and compute the first correction via a Malliavin/Wick (Nourdin–Peccati) expansion of the random connection root, using the program's Green's-function/chaos machinery.

Claim 7 (First stochastic Stokes correction)**.**

$$\Omega_2 = -0.451 + 0.352\,i, \qquad |\Omega_2| = 0.572, \quad \arg\Omega_2 \approx 142° \approx \tfrac{3\pi}{4}, \tag{8}$$

with an independent exact-Newton validation to 0.67%. [numerical]

Two structural payoffs follow, both consistent with the fluctuation-dominated picture of Bureković–Schäfer–Grauer (in which the Gaussian-fluctuation prefactor, not the classical action, sets a noise-induced tail):

(a) the $45° \to 50°$ shift is *not* a mean-shift effect at $O(\eta^2)$ (the connection-root mean rotates only $\approx -3.2°/\eta^2$, staying near $-45°$); it lives in the fluctuation sector (anisotropic, rms $|\delta\lambda| \approx 1.34\,\eta$). [numerical]

(b) the renormalization-clean object is $\mathbb{E}$[connection root], not the Borel *location* (whose mean shift is $\delta(0)$-divergent — the same threshold that first appears at v_3), because the log-derivative is a non-local Itô functional. [numerical]

6. The partial resurgent representation

Assembling Sections 3–5:

Claim 8 (Partial resurgent representation of $\mathcal{W}$)**.** The weak-noise expansion (3) of $\mathcal{W}_\beta$ is a resurgent trans-series in $\eta^2 = 4/\beta$ whose leading data are: the exact perturbative sector $v_0, \ldots, v_6$ (4), factorially divergent (Claim 4); a Borel plane with a complex-conjugate pair at $(\theta, |\zeta|) \approx (50°, 1.9)$ (6) and a competing real Freidlin–Wentzell instanton at $S = s^5/10$ (5); and a stochastic Stokes constant $\Omega = 1 + \eta^2\Omega_2 + \cdots$ with Ω_2 as in (8). [numerical] / [conjectural]

The representation is stated here and *tested* in Section 7; the honest ledger of what is solid, numerical, and open is deferred to Section 8, after the validation, so that it can account for the resummation evidence as well.

7. Validation: the resummed trans-series reconstructs $\mathcal{W}$

The strongest test of a resurgent representation is whether resumming it reproduces the true observable *outside* the perturbative radius. Write $x = \eta^2 = 4/\beta$ and $f(x) = \mathrm{Var}(Y^\star)/\eta^2 = \sum_{n\ge 0} v_n x^n$; the radius is $x_c \approx 1.2$, so $\beta = 2$ ($x = 2$) lies well outside it and the naive partial sums diverge there (-166 at seven terms). We resum by median/lateral Borel–Padé: form the Borel transform $B(u) = \sum_n (v_n/n!)u^n$, approximate it by the diagonal [3/3] Padé (all seven coefficients), and evaluate $f(x) = \int_0^\infty e^{-t} B(xt)\,dt$ along a contour rotated to $\pm\varphi$ to avoid the real instanton pole; the median $\frac{1}{2}(f_+ + f_-)$ is the physical (real) value. The comparison is against the Monte-Carlo-free FP-PDE ground truth, which reproduces the $\beta = 2$ fingerprint (skew $+0.601$, exkurt -0.243).

$x = \eta^2$	β	naive (7-term)	median resummation	ground truth
1.0	4.0	-3.2	0.224	0.232
1.5	2.7	-32.5	0.234	0.239
2.0	2.0	-166.4	**0.235**	**0.237**
2.25	1.8	-326.1	0.234	0.234

Table 2. Median Borel–Padé resummation of the seven-term series versus the MC-free FP-PDE ground truth for $f = \mathrm{Var}(Y^\star)/\eta^2$. At $\beta = 2$, well outside the radius, the resummation agrees to $\approx 1\%$ while the naive partial sum is wrong by orders of magnitude and in sign.

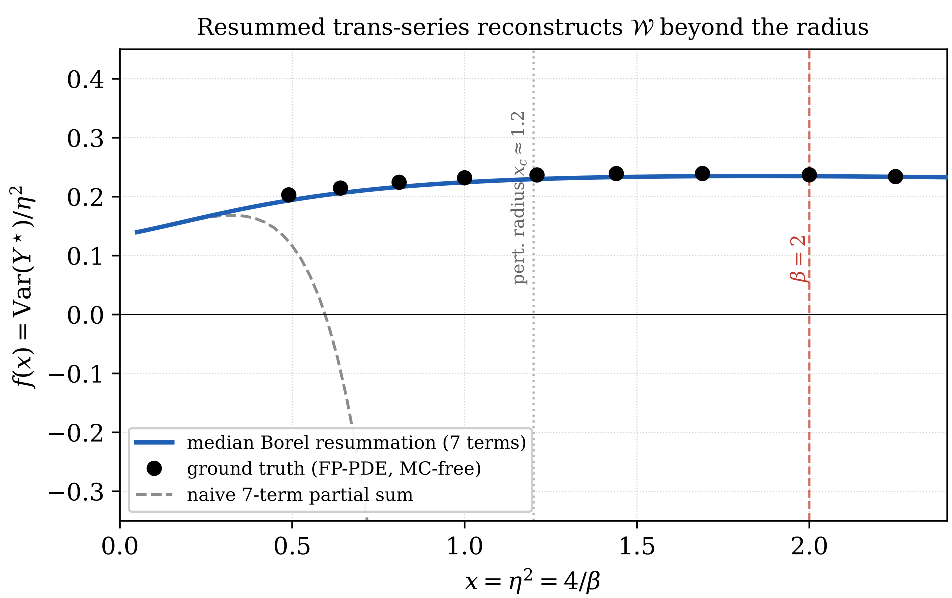

Figure 3. The median Borel resummation of the seven-term trans-series (solid) tracks the FP-PDE ground truth (dots) across the perturbative radius $x_c \approx 1.2$ to $\beta = 2$ and beyond, while the naive partial sum (dashed) diverges past $x \approx 0.6$.

Claim 9 (The resummation reconstructs $\mathcal{W}$ at $\beta = 2$)**.** The diagonal [3/3] median Borel–Padé resummation of $v_0, \ldots, v_6$ gives $f(x{=}2) = 0.235$ against the ground-truth 0.237 — agreement to $\approx 1\%$ at $\beta = 2$, outside the perturbative radius — and tracks the ground truth across $x \in [0, 2.4]$ (Table 2, Figure 3). [numerical]

Remark 10 (Honest caveat). With only seven coefficients the resummed value is sensitive to the Padé order: the best-conditioned diagonal [3/3] gives 1%, but off-diagonal orders scatter ([2/3] : 0.212, [3/2] : 0.203, [2/4] : 0.175, [4/2] fails). The $\approx 1\%$ headline is the diagonal approximant, the canonical choice; the scatter is the expected signature of a short series, and it will tighten with additional coefficients. The lateral-contour median is itself angle-independent (numerically exact). [numerical]

7.1. **A second cumulant: the κ_3 ladder and the skewness overshoot.** The same engine computes the third-cumulant weak-noise ladder, $\kappa_3(Y^\star) = \eta^4 \sum_{j\geq 0} t_j\, \eta^{2j}$ with $t_j = \sum_{a+b+c=4+2j} \mathrm{cum}_3(Y_a, Y_b, Y_c)$ (only even $a+b+c$ survive, by noise parity). The computed ladder $t_{0:5} = (0.059,\ 0.153,\ 0.191,\ -0.187,\ -2.07,\ -4.7)$ is again factorially divergent, so the resurgent structure is present in a *second* cumulant, not only the variance. Two checks pass. First, the leading skewness coefficient $t_0/v_0^{3/2} = 1.20$ agrees with the independently-known value ≈ 1.16. Second, the naive leading skewness at $\beta = 2$ is $\eta\, t_0/v_0^{3/2} = \sqrt{2} \cdot 1.20 \approx 1.7$ against the true 0.601 — the $\times 2.8$ "skewness overshoot" that first signalled $\beta = 2$ to be non-perturbative, here reproduced from the exact leading coefficient. [numerical]

Remark 11 (Skewness is a harder resummation). Unlike the variance, the six-term skewness series $S(x) = T(x)/V(x)^{3/2}$ (with $T = \sum t_j x^j$) is only marginally resummable: its coefficients grow much faster ($s_5 \sim 50$ against the variance's $v_6 \approx -1.9$), its top rungs t_4, t_5 are grid-sensitive, and a median Borel–Padé resummation at $\beta = 2$ scatters across Padé orders (0.4–2.2, versus the true 0.601). So the skewness *corroborates* the divergent/resurgent structure and reproduces the overshoot, but a quantitative $\beta = 2$ reconstruction of it would require more (and better-converged) coefficients; the variance (Section 7) remains the clean quantitative validation. [numerical]

8. Honest ledger

Solid/exact: the two obstructions (Props. 1–2); the deterministic connection root λ_0; the coefficients v_0–v_4 (exact, Monte-Carlo validated); the instanton exponent 5; the $\frac{1}{2}$-boundary convention; factorial divergence. *Solid, grid-extrapolated:* v_5, v_6. *Numerically established:* $(\theta, |\zeta|) \approx (50^\circ, 1.9)$; Ω_2; the resummation reconstructing $\mathcal{W}$ at $\beta = 2$ to $\approx 1\%$ (Claim 9); the κ_3 ladder and the leading skewness / $\times 2.8$ overshoot (Section 7.1). *Corroborating but not clean:* the full skewness resummation (Remark 11). *Conjectural:* that the leading data above determine the full trans-series; the precise inheritance relation between λ_0 and the stochastic pair. *Open frontier (mapped):* a *proved* stochastic exact-WKB (Section 9). We emphasize what the coefficients could not settle: a converged v_7 — the sharp 45°-vs-50° discriminator (pre-registered $v_7 \approx +1.3$ at 45° vs +8.2 at 50°) and the datum that separates the real instanton from the pair — remains out of computational reach on a single machine, and $\mathcal{W}$ is finalized here at v_6.

9. The mapped frontier: proved stochastic exact-WKB

The one load-bearing blank is a rigorous exact-WKB / resurgence theory for the *noise-dressed* Weber operator: no random-potential edge law has been given a proved resurgent trans-series, and $\mathcal{W}$ would be first-of-kind. The scoped route is the algebraic theory of parametric Stokes phenomena (Aniceto–Crew), which rigorously encodes Borel singularities that move on a parameter-dependent algebraic curve and can cross the integration contour — the exact mechanism a noise-shifted Borel phase requires; the genuinely novel step is promoting the deformation parameter to the noise coupling, i.e. making the Voros symbols random variables on a fixed deterministic curve. The minimal target lemma is to show that averaging the random Voros symbol over the noise fixes the Borel-singularity *locations* while shifting only the Stokes *constant*, thereby defining Ω rigorously as $\mathbb{E}$[random Voros datum] and reducing the frontier to the computation already begun in Section 5. The remaining bridge is to match Ω_2 to Hao's deterministic residue — is the stochastic Stokes constant $1 + \eta^2(\cdots)$ in Hao's normalization? [conjectural]

Acknowledgements

Computational tools: `fp_cusp.py` (Monte-Carlo-free ground truth), `chaos_diagram.py/chaos_transfer.py` (exact Wiener-chaos coefficients), `connection_closed_form.py` (deterministic connection), `instanton_action.py` (left-tail instanton), `stochastic_stokes_o_eta2.py` (Ω_2).

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